

Summary Report: SDR Platforms and Spectrum Monitoring

1. Comparative Analysis of SDR Devices

Objective: To select an SDR platform following a phased approach (Low-cost → Scalable) while minimizing *vendor lock-in* through a portable processing pipeline: IQ capture → PSD/Waterfall → Metadata/Storage.

SDRs Considered: RTL-SDR (RX), Adalm-Pluto (TX/RX), HackRF One (TX/RX), and USRP N210 (Ethernet-based).

Device	Resolution	Bandwidth	Key Strengths & Use Cases
RTL-SDR	8-bit	~2.4 MHz	Extremely low cost. Ideal for Phase 0: signal inventory, basic FM/AM monitoring, and IQ-to-PSD flow validation.
PlutoSDR	12-bit	Up to 56 MHz	Full-duplex (AD936x). Excellent for Phase 1: portable lab prototyping, FPGA-based processing, and IIO ecosystem.
HackRF One	8-bit	20 MHz	Wide range (1 MHz–6 GHz). Half-duplex. Best for Phase 1: wideband sweeps and rapid RF exploration across multiple bands.
USRP N210	14-bit	Up to 50 MHz	Industrial grade. Ethernet interface provides high stability and repeatability. Phase 2: high-performance monitoring and modular scaling.

Technical Comparison and Trade-offs

- **Dynamic Range:** The 8-bit architecture of the RTL-SDR and HackRF limits the Effective Number of Bits (ENOB), making them susceptible to saturation in the presence of strong nearby signals (e.g., LTE or FM towers). The 12/14-bit resolution of Pluto and USRP provides a much higher dynamic range, essential for detecting weak signals near the noise floor.
- **Frequency Stability:** Most low-cost SDRs suffer from frequency drift as they heat up. The USRP N210 and newer Pluto revisions include or support high-stability TCXOs or external 10 MHz references, which are critical for long-term automated monitoring.
- **Data Throughput:** USB 2.0 (RTL-SDR/HackRF) creates a bottleneck for wideband real-time analysis. The USRP's Gigabit Ethernet interface allows for more reliable high-rate streaming and easier remote deployment via network infrastructure.

Scope and Use Cases

Spectrum monitoring ranges from simple *sweeps* (occupancy detection) to detailed *characterization* (bandwidth, relative power, temporal persistence, and basic classification).

- **Signal Inventory by Band:** Identifying persistent carriers, intermittent signals, and hourly usage patterns.

- **Noise Floor Detection:** Defining robust thresholds based on smoothed PSD and statistical averages (mean/median).
- **IQ Recording with Metadata:** Generating datasets for offline analysis (classification, anomaly detection) while maintaining full traceability of capture parameters.

Selection Criteria (Modularity and Scalability)

- **Pipeline Portability:** Prioritize software abstractions (like SoapySDR or GNU Radio) that allow hardware swapping without major code rewrites.
- **RF Environment Adaptability:** In dense environments, 8-bit SDRs require external pre-selection (filters) and attenuation to prevent front-end overload.
- **Synchronization:** If the scope expands to TDOA (Time Difference of Arrival) or AoA (Angle of Arrival), the hardware must support external clock/PPS distribution.
- **Maintainability:** Documenting gains (LNA, VGA, Mixer), sample rates (F_s), and antenna types to ensure measurements are repeatable across different campaigns.

Recommended Phased Roadmap

- 1) **Phase 0 (Exploration):** Deploy RTL-SDR for initial site surveys and validating the basic software logic for IQ processing and database storage.
- 2) **Phase 1 (Prototyping):** Use PlutoSDR or HackRF for controlled TX/RX testing and generating training datasets for signal classifiers.
- 3) **Phase 2 (Scaling):** Transition to USRP N210 for high-fidelity monitoring where repeatability, modular frequency coverage (via daughterboards), and network-based scaling are required.

Hardware Integration and Best Practices

To avoid manufacturer lock-in, the system should be decoupled into discrete blocks:

1. **RF Front-end:** Antennas (Discone for VHF/UHF, Horn for Microwaves) and external filtering/amplification.
2. **SDR Layer:** Utilizing standard drivers (UHD, Libiio) or abstraction layers (SoapySDR).
3. **Processing Layer:** Decoupled IQ data storage using formats like SigMF (Signal Metadata Format) to record center frequency, gain, and timestamps.

Expected Limitations and Mitigations

- **Strong Signal Saturation:** Mitigated by using notch filters (e.g., FM bandstop) and manual gain control instead of AGC.
- **Measurement Variability:** Controlled by using fixed gain settings and performing periodic calibration against a known reference.
- **Frequency Offset:** Mitigated by using GPSDO (GPS Disciplined Oscillators) or high-quality internal TCXOs where precision is paramount.